

Project Report Format

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INTRODUCTION

1.1 Project Overview

The project “Empowering India: Analysing the Evolution of Union Budget Allocations for Sustainable Growth” focuses on a detailed analysis of India’s Union Budget allocations for the financial years 2021–22 to 2023–24. The Union Budget is a key financial document presented annually by the Government of India, outlining revenue generation and expenditure plans aimed at economic development and social welfare. This project studies how budget allocations across various ministries, departments, and schemes have evolved over time, especially in response to challenges such as the COVID-19 pandemic and the need for sustainable and inclusive growth.

The project involves collecting and storing budget data in a database, performing SQL operations, and using Tableau for data preparation, visualization, dashboards, and storytelling. Multiple visualizations are created to identify trends, compare allocations across years, and highlight priority sectors. The insights generated are integrated into a web interface using Flask, enabling interactive exploration of dashboards and stories. Overall, the project provides a data-driven view of government spending patterns and their role in supporting India’s economic resilience and long-term growth

1.2 Purpose

The main purpose of this project is to analyze and visualize the changes in Union Budget allocations over recent financial years to understand government priorities and their impact on sustainable development. By examining budget data across different categories, ministries, and schemes, the project aims to uncover meaningful insights that support informed decision-making for businesses, policymakers, and researchers.

Additionally, the project seeks to demonstrate the effective use of SQL, Tableau, and web integration technologies in transforming raw financial data into interactive dashboards and data stories. Through clear visualizations and performance-optimized dashboards, the project helps users easily interpret complex budget data, identify trends, and assess how public funds are allocated to support economic recovery, innovation, infrastructure, healthcare, and other key sectors contributing to India’s growth.

IDEATION PHASE

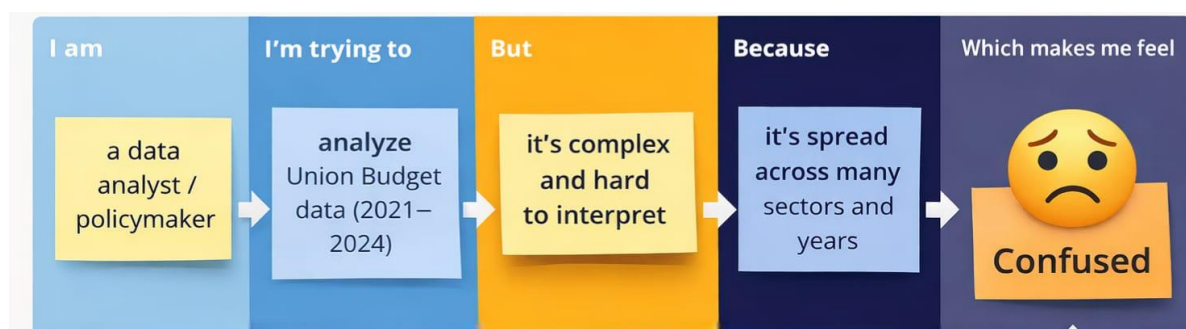
2.1 Problem Statement

Customer Problem Statement Template:

Stakeholders such as policymakers, analysts, and researchers need a clear and structured way to understand Union Budget allocations from 2021 to 2024 because the data is complex and spread across multiple sectors and schemes. Without proper analysis and visualization, it is difficult to identify priority areas, track budget trends, and evaluate government spending effectiveness. Aligning budget data with visual insights helps stakeholders make informed decisions and assess sustainable growth initiatives.

I am	An Indian citizen who is a taxpayer, middle-class, and interested in India's economic policies
I'm trying to	Understand the proposed budget and how it will affect me and the Indian economy
but	The budget details can be complex, filled with technical jargon, and difficult to interpret
because	There is a lack of clear information and simplified explanations available
which makes me feel	Confused, concerned, and uncertain about the financial implications of the budget
	Confused, concerned, and uncertain about the financial implications of the budget

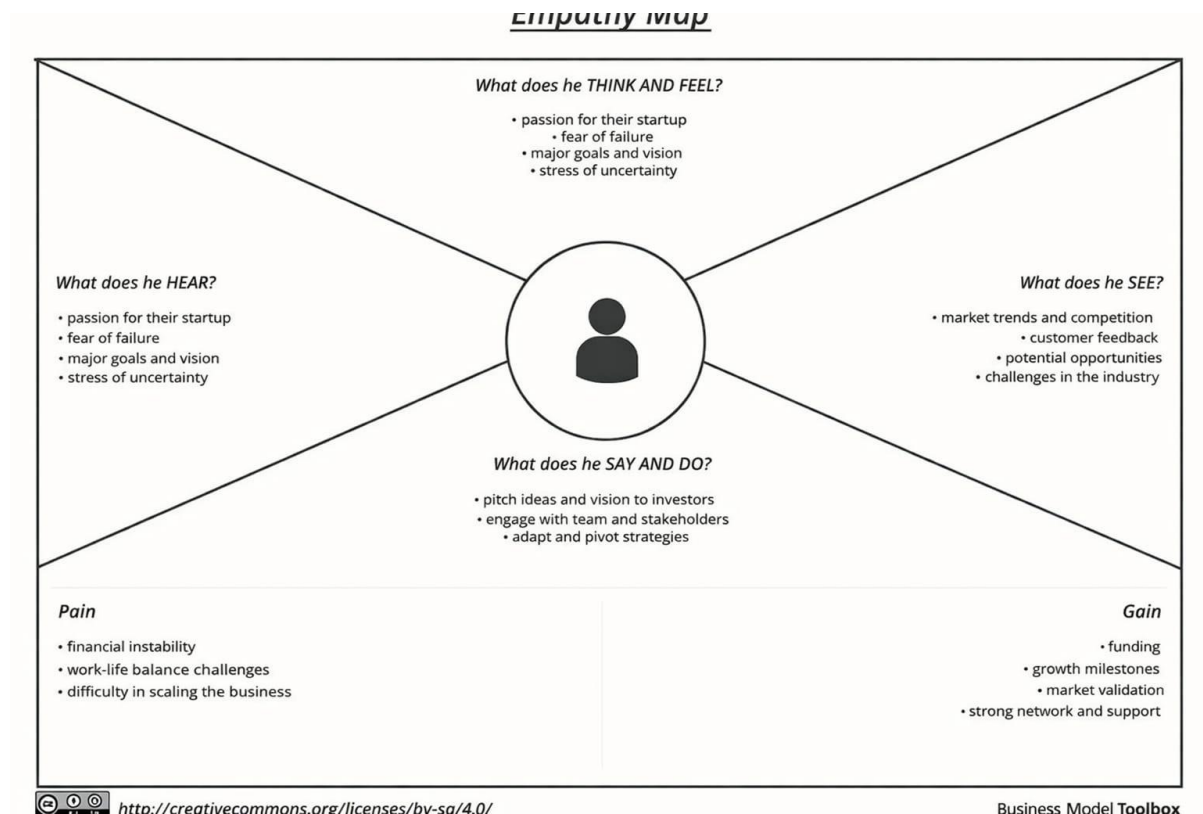
Example:



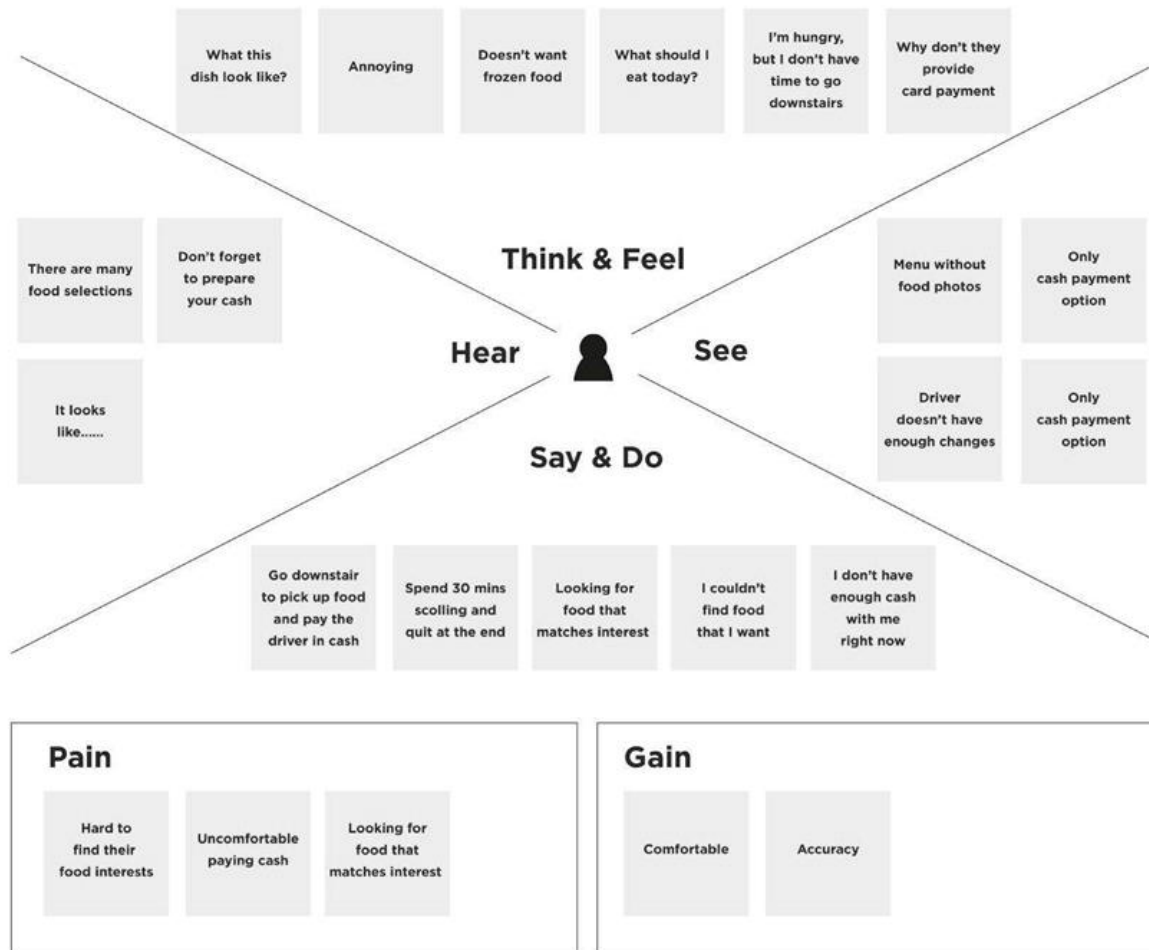
2.2 Empathy Map Canvas

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A data analyst / researcher	analyse Union Budget allocations from 2021–2024	the budget data is complex	it contains large volumes of numerical data across multiple sectors	confused and uncertain while identifying clear trends

The users of this project, mainly Indian citizens, think about how the Union Budget affects their daily life, income, taxes, and future opportunities, but they often feel confused and uncertain due to complex budget language. They see rising prices, mixed media opinions, and government claims, while hearing conflicting views from news channels, experts, and society. These factors create pain points such as lack of clarity, fear of inflation, and difficulty understanding real benefits or losses from the budget. This project helps users gain clear, data-driven insights into budget priorities, enabling better financial awareness, informed decisions, and trust in transparent analysis.



Example: Food Ordering & Delivery Application



Brainstorm & Idea Prioritization:

This session focuses on exploring ideas to analyse India's Union Budget data from 2021 to 2024. Team members brainstorm key sectors, schemes, and ministries based on budget allocations. Ideas are discussed to identify spending trends and priority areas. The most impactful and relevant insights are shortlisted for visualization. These prioritized ideas help in building clear dashboards and meaningful data stories.

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Union Budget Analysis 2021 - 2024

Analyzing India's Union Budget allocations from fiscal years 2021-2024 to uncover insights into government spending across various sectors and ministries. Connecting the data with Tableau to visualize budget trends and perform comprehensive analysis.

- Connect DB with Tableau
- ✓ Create dashboards & Stories
- ✓ Analyze budget trends



1 Data Collection & Tableau

Collecting and preparing relevant budget allocation data for analysis in Tableau to begin visualizing trends and insights.

10 minutes

Data Collection

- > Collect the finalized dataset ensuring it is clean, relevant, and aligned with the defined problem.

Open article →

Data Preparation

- > Clean, transform, and organize the data to ensure consistency and accuracy before analysis in Tableau.

Open article →

Initial Visualizations

- > Begin creating initial visualizations to uncover trends in budget allocations.

2 Dashboard & Story

Creating dynamic Tableau dashboards and data stories to illustrate budget allocations and trends. Embedding them into a Flask-based web application for interactive insights.

1 hour

Performance Testing

- > Optimize dashboard performance to handle queries effectively with large datasets.

Web Integration with Flask

- > Embed dashboards and interactive stories in a Flask-based web application for seamless access.



Step-2: Brainstorm, Idea Listing and Grouping

1 Brainstorm

Write down any ideas that come to mind that address the project's theme "Empowering India: Analysing the Evolution of Union Budget Allocations for Sustainable Growth"

10 minutes

TIP You can select a sticky note and hit the 'detach' switch to quickly move it to other clusters. Must be set on 'locking' as well.

Tamarana Shruthi	Thadicherla Chandra Manikanta	Sri Devi Prasad
Incentives for EV manufacturers	Govt support for renewable energy	Funds for COVID resilience
Green energy subsidies	Funding for EV charging infrastructure	Expansion funds for EV industry
Funds for renewable research	Healthcare innovation grants	Budget focus on R&D industry
Support for electric buses	Fiscal support to boost EV industry	Incentive schemes existing

Allaka Venkateswara Rao	Ananthapalli Rohitha	Balla Ramadevi
Tax rebates for EV production	Tax rebates for EV production	Priority for startups development
Increase incentives for R&D	Funding for AI/ML sector	Protecting electronic executives
Expansion funds for pharma sector	Pratiyeka or to in financial reserves	Boosting for new drug development
Funds for new drug development	Stimulating us protects technology-based-pharma sectors	Scaling for new drug development

Allaka Venkateswara Rao	Ananthapalli Rohitha	Balla Ramadevi
Tax rebates for pharma industry	Budget for renewable energy projects	Budget for renewable startups
Increase incentives for pharma	Startups in green tech	Central startups

2 Group Ideas

Take turns sharing your ideas while clustering similar or related notes as you go. In the last 10 minutes, give each cluster a sentence title label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub groups.

20 minutes

TIP Add customization tags to sticky notes for better organization: color-coded, numbered, grouped, etc. to make it easier to manage.

Group Ideas

- ✓ **EV & Renewable Energy Opportunities**
 - Govt support for renewable energy
 - Green energy subsidies
 - Funding for EV charging infrastructure
 - Funding for EV manufacturers
 - Incentives for EV manufacturers
 - Incentives funds for EV research startups
 - Expansion funds for EV industry
 - Repensation aid to benefits
 - Renewables ID and pharma sector
- ✓ **Healthcare & Pharmaceutical Initiatives**
 - Budget focus on sustainable startups
 - Talents break COVID resilience
 - Funds for maros profitable startups
 - Budget incentives for startups
 - Support for Startups
 - Central schemes for startups
 - Boosting R&D in pharma sector
 - Grants for development
- ✓ **Support for & Development Incentives**
 - Increase incentives for R&D
 - Budget focus on R&D in pharma
 - Boosting R&D in pharma R&D
 - Boosting R&D in pharma
 - Grants for development
- ✓ **Research & Development Incentives**
 - Increase incentives for R&D
 - Boosting R&D in pharma sector
 - Boosting for pharma R&D
 - Tax incentives for pharma R&D

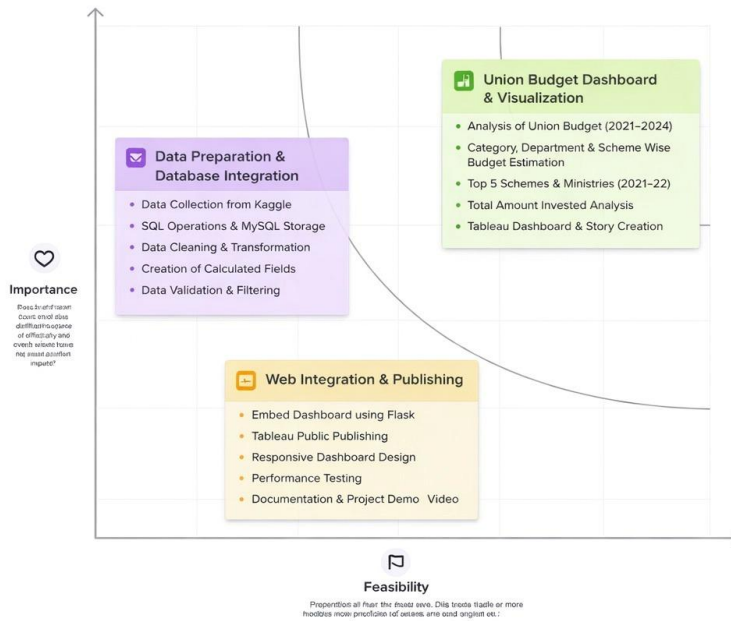
Step-3: Idea Prioritization

1

Prioritize

Your team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

20 minutes



REQUIREMENT ANALYSIS

3.1 Customer Journey map



3.2 Solution Requirement

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should provide a user-friendly and intuitive interface with easy navigation, clear labels, responsive dashboard design and interactive visualizations so that users can easily analyze Union Budget data without technical expertise.
NFR-2	Security	The application must ensure secure user authentication and authorization. Sensitive data must be protected using encryption (HTTPS), secure login mechanisms, and role-based access control to prevent unauthorized access.
NFR-3	Reliability	The system should operate consistently without failures and ensure accurate data processing. It must handle errors gracefully and maintain data integrity during data loading, analysis, and visualization.
NFR-4	Performance	The dashboard and visualizations should load within 3-5 seconds for standard datasets. SQL queries and Tableau integrations must be optimized to ensure smooth performance even with large budget data.
NFR-5	Availability	The system should be available 24/7 with minimal downtime. In case of maintenance, proper notifications should be provided. Hosting and deployment should ensure high uptime.
NFR-6	Scalability	The system must support future expansion, such as adding more years of budget data, additional ministries, or new visualization modules without affecting performance.

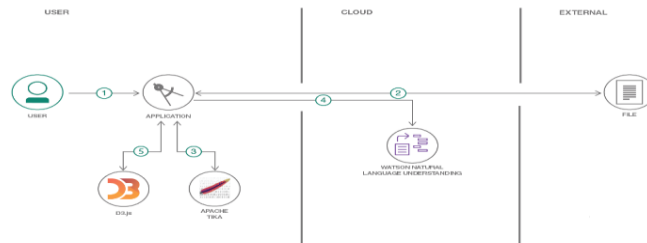
3.3 Data Flow Diagram

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example:

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

User Stories

Use the below template to list all the user stories for the product.

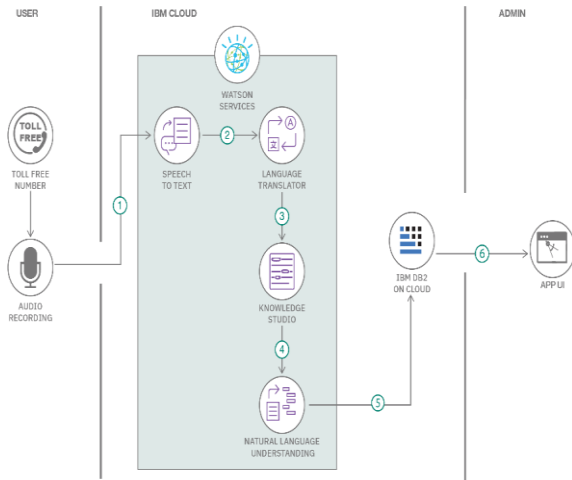
User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my	I can access my account / dashboard	High	Sprint-1
		USN-2	As a user, I will receive confirmation email once I have registered for the application	I can receive confirmation email & click confirm	High	Sprint-1
		USN-3	As a user, I can register for the application through Facebook	I can register & access dashboard with Facebook	Low	Sprint-2
		USN-4	As a user, I can register for the application through Gmail	I can receive confirmation email	Medium	Sprint-1
Customer (Web user)	Login	USN-5	As a user, I can log into the application by entering email & password	I can log in to the application dashboard	High	Sprint-1
Customer Care Executive	Dashboard	USN-6	As a user, I can view account summary, recent transactions, and budget analytics in the dashboard	I can see account summary recent transactions-and analytics in the dashboard	High	Sprint-2
Customer Care Executive	Dashboard	USN-7	As a customer care executive, I can resolving user queries and support tickets within the dashboard	I can view and respond to user queries and manage support tickets within dashboard	Medium	Sprint-2
Administrator	Dashboard	USN-8	As an administrator, I can view user activity logs and system settings within the dashboard	I can view user activity logs and adjust system settings within the dashboard	High	Sprint-2

3.4 Technology Stack

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services

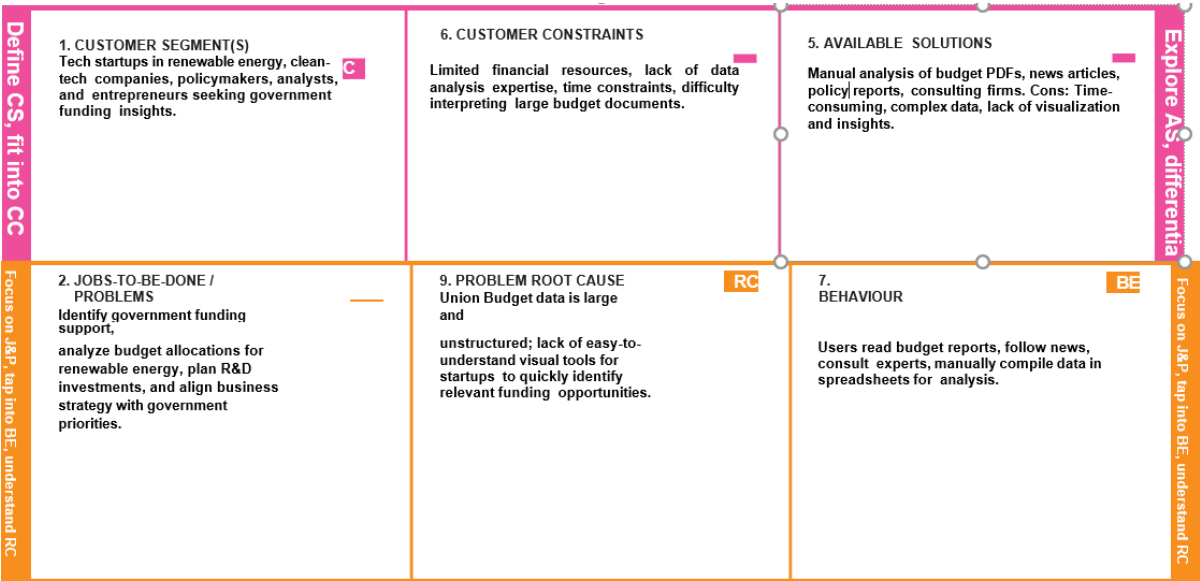
Description	Technology
How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, JavaScript / Angular JS / React JS etc.
Logic for a process in the application	Java / Python
Logic for a process in the application	IBM Watson STT service
Logic for a process in the application	IBM Watson Assistant
Data Type, Configurations etc.	MySQL, NoSQL, etc.
Database Service on Cloud	IBM DB2, IBM Cloudant etc.
File storage requirements	IBM Block Storage or Other Storage Service or Local Filesystem
Purpose of External API used in the application	IBM Weather API, etc.
Purpose of External API used in the application	Aadhar API, etc.
Purpose of Machine Learning Model	Object Recognition Model, etc.
Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration:	Local, Cloud Foundry, Kubernetes, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The application uses open-source technologies for data processing, visualization, and web integration to ensure cost-effectiveness and flexibility.	Python, Flask, MySQL, Tableau Public
2.	Security Implementations	The system ensures secure access through role-based authentication and encrypted communication. Data validation and secure database connections are implemented to prevent unauthorized access.	HTTPS, Role-Based Access Control (RBAC), SHA-256 Password Hashing, OWASP Best Practices
3.	Scalable Architecture	The application follows a 3-tier architecture (Presentation Layer, Application Layer, Database Layer). It supports future expansion such as adding more years of budget data or additional dashboards.	Flask (Backend), Tableau (Visualization Layer), MySQL (Database)
4.	Availability	The system ensures high availability by hosting the dashboard online and maintaining reliable database connectivity. Backup mechanisms ensure data safety.	Tableau Public Hosting, Cloud Deployment, Database Backup Mechanism
5.	Performance	The system is optimized for fast query execution and dashboard loading. SQL indexing and efficient filtering improve response time even with large datasets.	MySQL Indexing, Optimized SQL Queries, Tableau Data Extracts

PROJECT DESIGN

4.1 Problem Solution Fit



3. Launching new R&D initiatives, funding shortages, new government budget announcements, policy changes in energy and sustainability sectors. TR	10. YOUR An interactive Tableau-based dashboard system that visualizes Union Budget allocations (2021-2024) for renewable energy-related ministries and schemes, embedded in a web app using Flask for easy access and decision-making. SL	8. CHANNELS of Online: Government budget websites, news portals, analytics blogs. Offline: Business meetings, policy forums, startup CH
4. EMOTIONS: BEFORE / AFTER Before: Uncertain, confused, worried about funding. After: Informed, confident, strategically prepared. EM		

4.2 Proposed Solution

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Government Union Budget data is available in large reports and PDFs, which makes it difficult for students, startups, and organizations to clearly understand budget allocation trends across ministries, schemes, and sectors such as renewable energy, EVs, and healthcare.
2.	Idea / Solution description	Develop an interactive data analytics solution using Tableau dashboards and stories , connected to a database, to visualize Union Budget allocations (2021–2024) category-wise, department-wise, and scheme-wise. The dashboards are embedded into a Flask-based web application to allow easy access and exploration of budget insights.
3.	Novelty / Uniqueness	Instead of static reports, this solution provides interactive visual dashboards and data stories that allow users to filter, compare, and analyze Union Budget trends over multiple years in one place, making complex financial data simple and user-friendly.
4.	Social Impact / Customer Satisfaction	This project improves public awareness and transparency of government spending. It helps startups, students, and policymakers make informed decisions by understanding funding priorities, which can support innovation in renewable energy, EVs, and healthcare.
5.	Business Model (Revenue Model)	The platform can follow a freemium model where basic dashboards are free for public use, and advanced analytics, custom reports, or institutional access can be offered to organizations, startups, and research institutions on a subscription or service-based model.
6.	Scalability of the Solution	The solution is scalable as new budget years, additional datasets (state budgets, sector-wise reports), and more dashboards can be added easily. The web-based deployment allows the platform to support multiple users and future feature expansions.

4.3 Solution Architecture

Solution Architecture:

The solution architecture illustrates how Union Budget data is transformed into meaningful insights through a layered system. The architecture connects raw government budget data with visualization tools and a web application to help users understand budget allocations easily.

- 1. Data Source Layer:**

Union Budget datasets (2021–2024) are collected from official sources/Kaggle. This layer acts as the input for the system.

- 2. Data Storage Layer (MySQL Database):**

The collected data is cleaned, structured, and stored in a MySQL database. SQL queries are used to retrieve and process the required information for analysis.

- 3. Visualization Layer (Tableau Dashboards & Stories):**

Tableau is connected to the database to create interactive dashboards and stories. These visualizations present budget allocations category-wise, department-wise, and scheme-wise, making complex financial data easy to interpret.

- 4. Web Application Layer (Flask):**

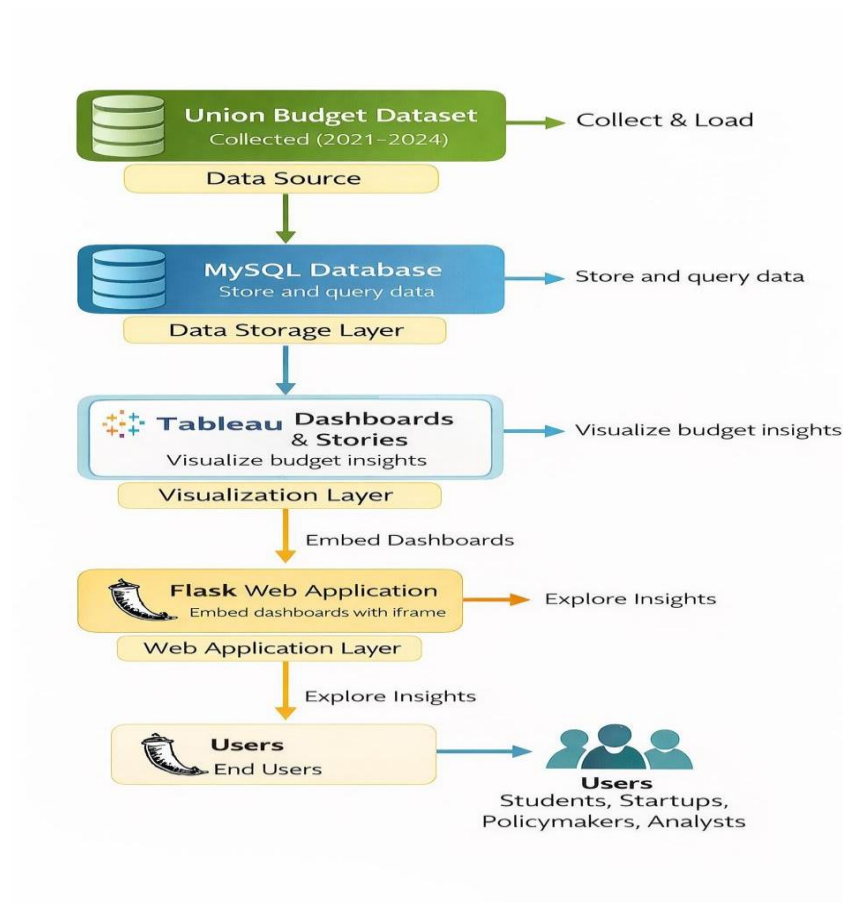
The Tableau dashboards are embedded into a Flask-based web application using iframes. This layer provides a simple user interface for interacting with the visualizations through a web browser.

- 5. End Users:**

Students, startups, policymakers, and analysts access the web application to explore budget trends, compare allocations, and gain insights for decision-making.

Solution Architecture Diagram:

Figure 1: Architecture and data flow of the *Empowering India*



PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint Timeline – Indian Union Budget (2001–2014)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	10	7 Days	01 Feb 2025	07 Feb 2025	10	07 Feb 2025
Sprint-2	8	7 Days	10 Feb 2025	16 Feb 2025	8	16 Feb 2025
Sprint-3	10	7 Days	18 Feb 2025	24 Feb 2025	9	25 Feb 2025
Sprint-4	6	7 Days	27 Feb 2025	05 Mar 2025	6	05 Mar 2025

Sprint Timeline – Indian Union Budget (2001–2014)

- **Sprint-1:** Data Collection, Storage & Cleaning
- **Sprint-2:** Budget & Scheme Analysis
- **Sprint-3:** Dashboard Development & Web Integration
- **Sprint-4:** Deployment & Testing

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

FUNCTIONAL AND PERFORMANCE TESTING

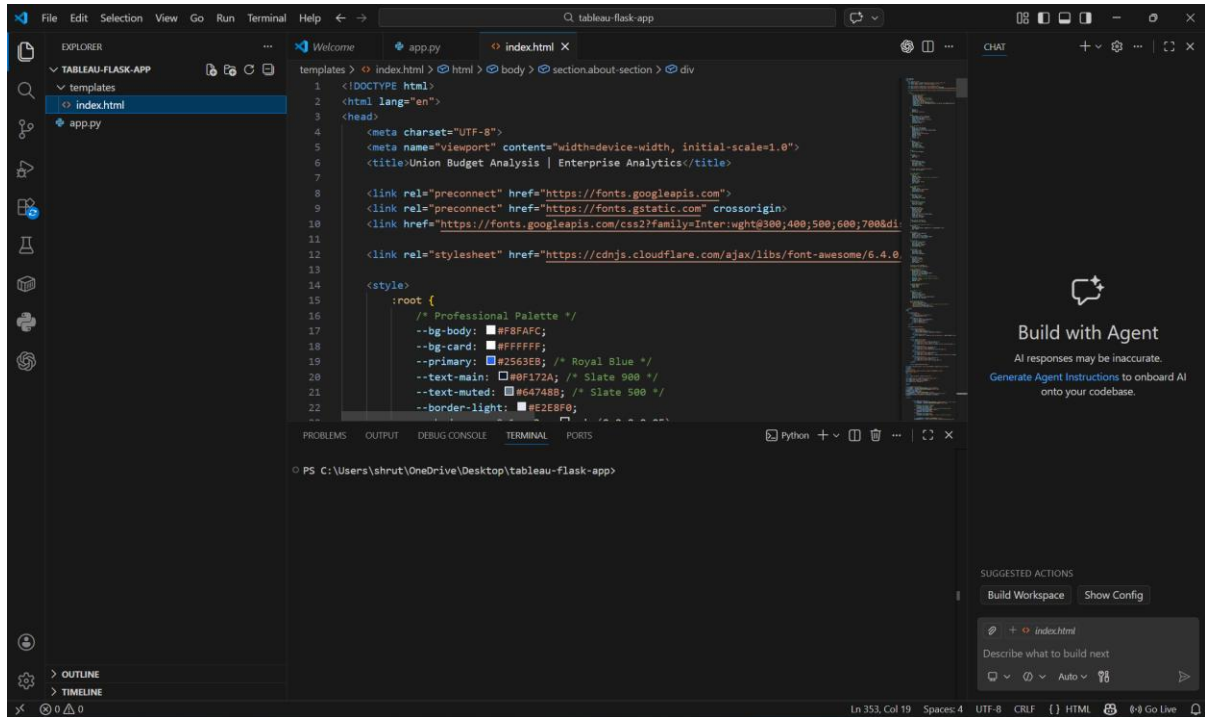
6.1 Performance Testing

Model Performance Testing

Project team has filled the following information for dashboard performance and analytical validation.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Dataset: Indian Union Budget (2001–2014) • Total Years: 14 • Ministries Covered: 50+ • Schemes: 200+
2.	Data Preprocessing	• Removed null values • Standardized column names • Converted financial values to numeric format • Verified year-wise consistency
3.	Utilization of Filters	• Year Filter • Ministry Filter • Scheme Category Filter • Allocation Range Filter
4.	Calculation Fields Used	• Total Allocation (SUM) • Year-wise Growth % • Ministry-wise Budget Share% • Top 5 Scheme Ranking
5.	Dashboard Design	No. of Visualizations / Graphs – 8 (Bar Chart, Line Chart, Pie Chart, Trend Analysis, KPI Cards)
6.	Story Design	No. of Story Pages – 4 (Overview, Ministry Analysis, Scheme Analysis, Year Comparison)

RESULTS



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Union Budget Analysis | Enterprise Analytics</title>
7
8   <link rel="preconnect" href="https://fonts.googleapis.com">
9   <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
10  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&dis
11
12  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0
13
14  <style>
15    :root {
16      /* Professional Palette */
17      --bg-body: #F8FAFC;
18      --bg-card: #FFFFFF;
19      --primary: #2563EB; /* Royal Blue */
20      --text-main: #0F172A; /* Slate 900 */
21      --text-muted: #64748B; /* Slate 500 */
22      --border-light: #E2E8F0;
23    }
24  
```

Build with Agent

AI responses may be inaccurate.

Generate Agent Instructions to onboard AI onto your codebase.

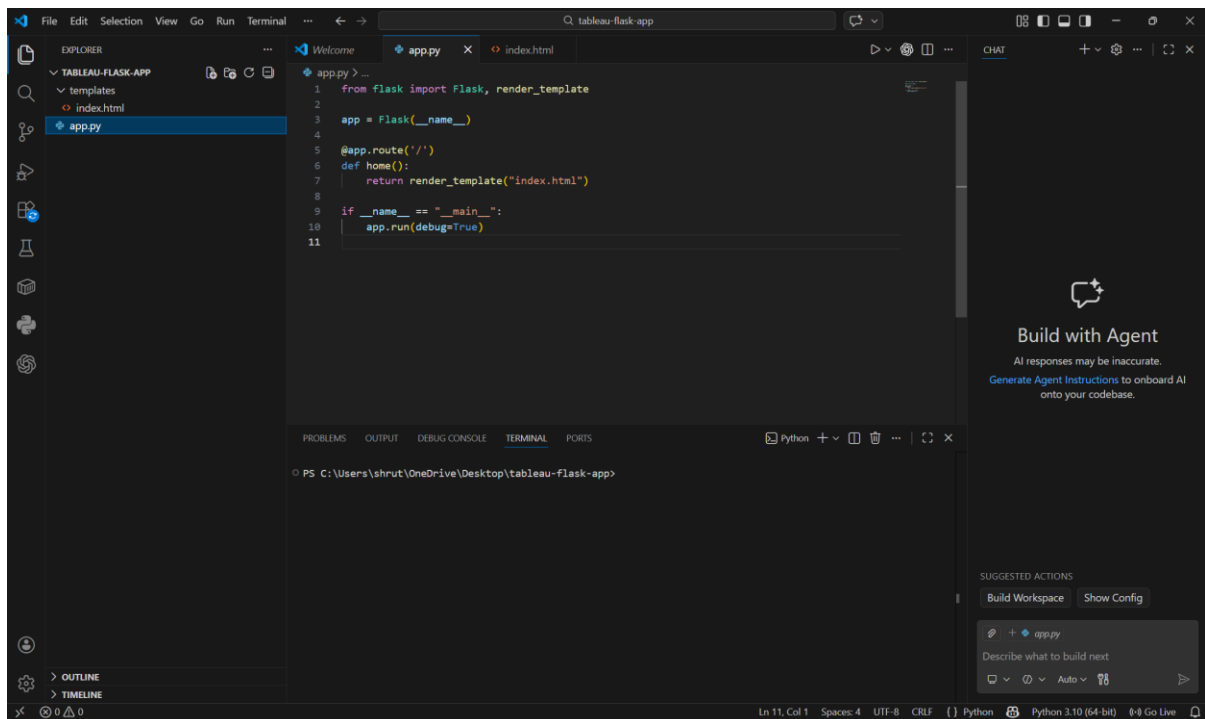
SUGGESTED ACTIONS

Build Workspace Show Config

Describe what to build next

Python

Ln 353, Col 19 Spaces: 4 UTF-8 CRLF {} HTML 8:8 Go Live



```
1 from flask import Flask, render_template
2
3 app = Flask(__name__)
4
5 @app.route('/')
6 def home():
7     return render_template("index.html")
8
9 if __name__ == "__main__":
10     app.run(debug=True)
11
```

Build with Agent

AI responses may be inaccurate.

Generate Agent Instructions to onboard AI onto your codebase.

SUGGESTED ACTIONS

Build Workspace Show Config

Describe what to build next

Python

Ln 11, Col 1 Spaces: 4 UTF-8 CRLF {} Python Python 3.10 (64-bit) 8:8 Go Live

About This Project

Academic Data Analytics Portfolio Project

Project Overview

This platform provides a visual exploration of India's fiscal allocations. It is designed to transform complex financial data into accessible insights, analyzing sector-wise distribution, expenditure trends, and historical budget shifts.

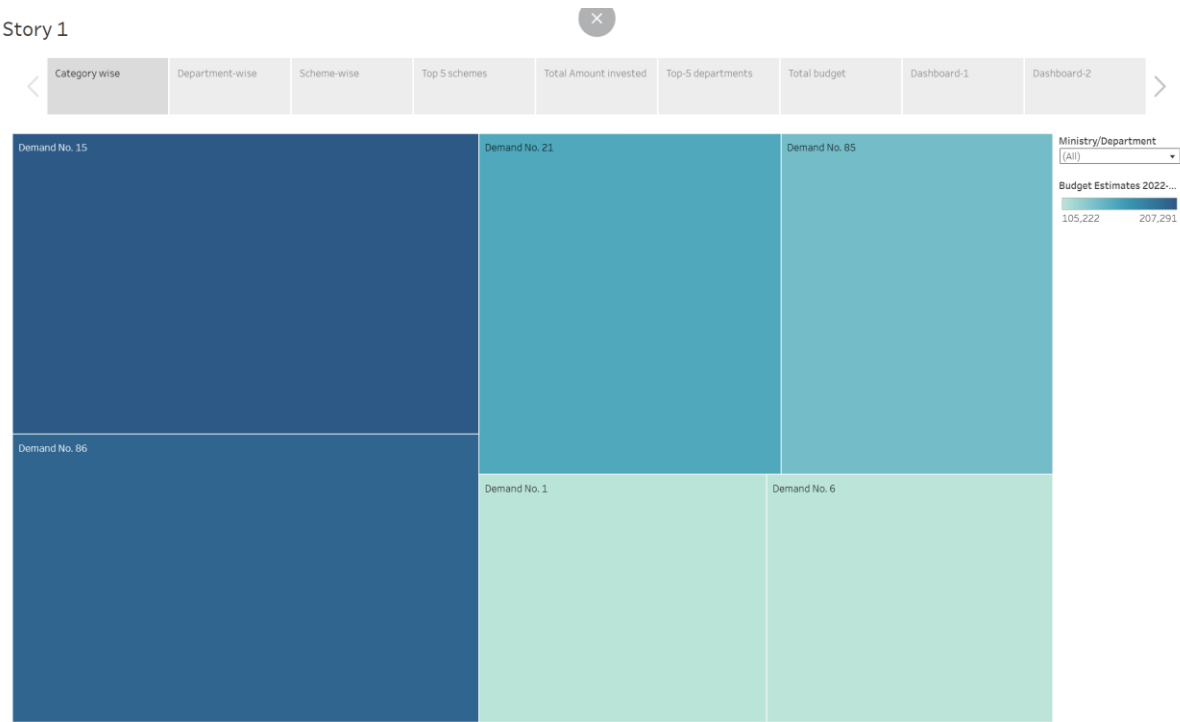
Data & Architecture

Data is sourced from official Ministry of Finance publications, cleaned via Python (Pandas), and visualized using Tableau Public.

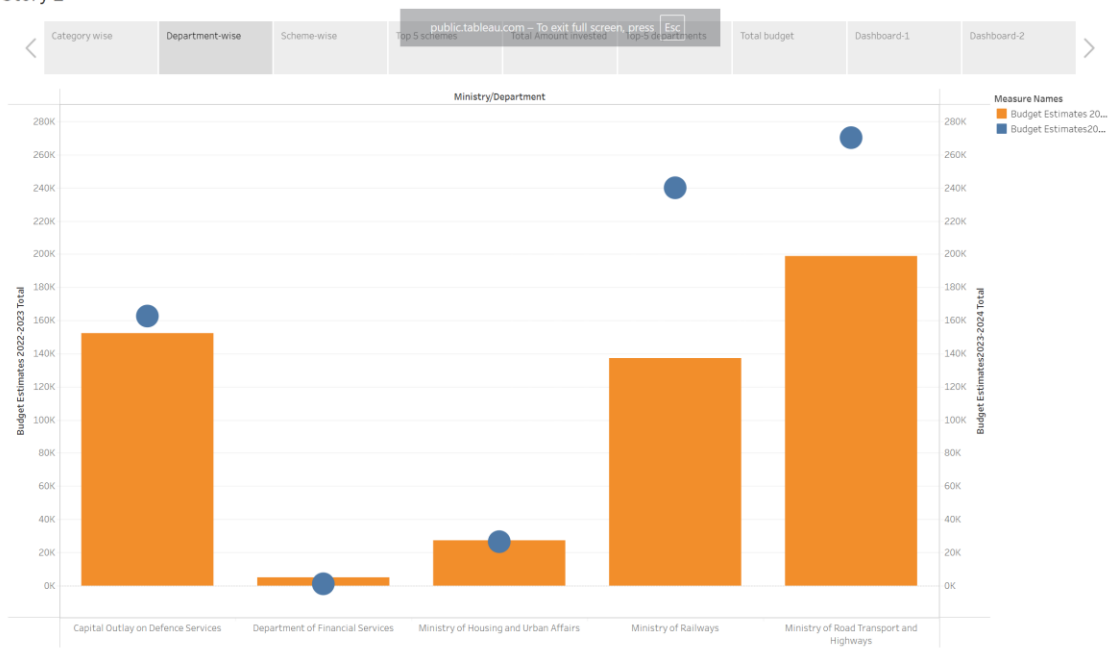
Flask Tableau Python/Pandas

© 2024 BudgetAnalytics Project. Data sourced from Union Budget of India.

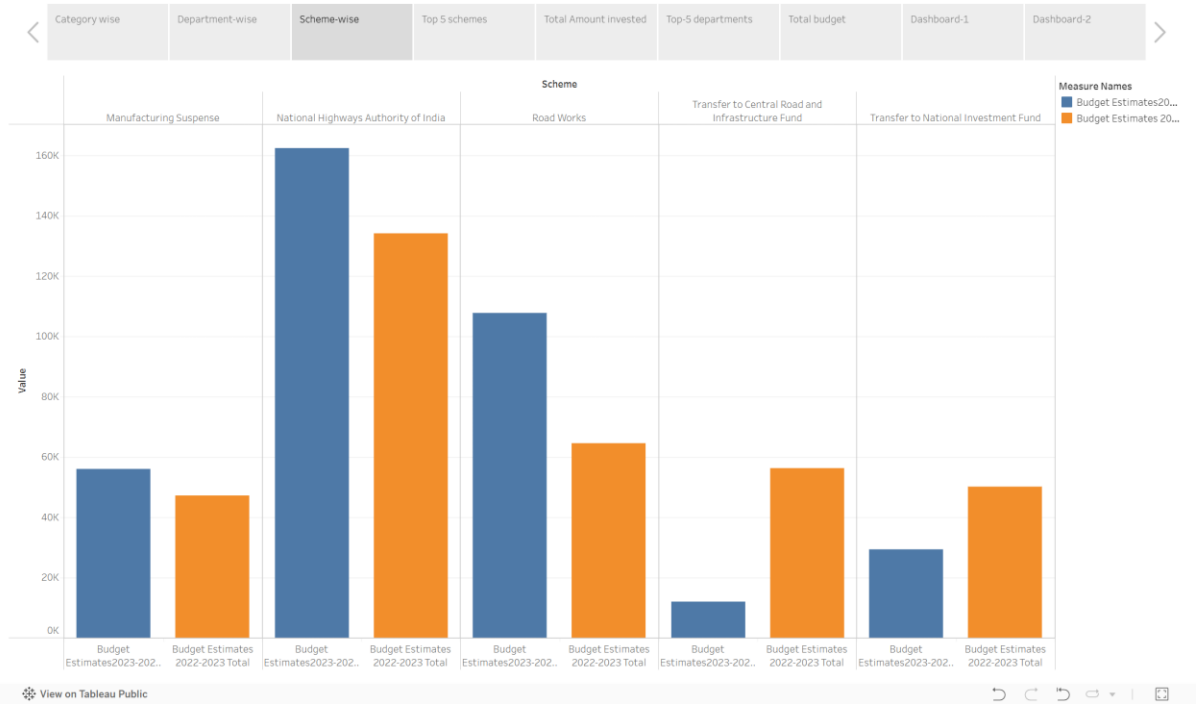
Story 1



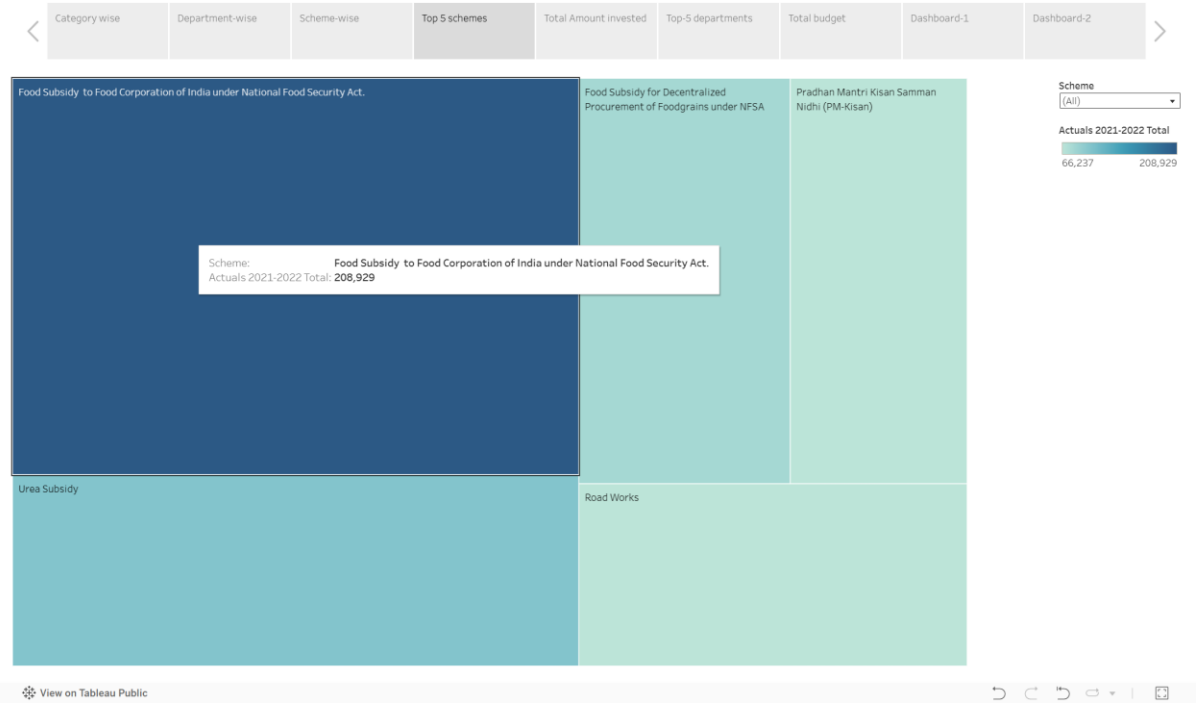
Story 1



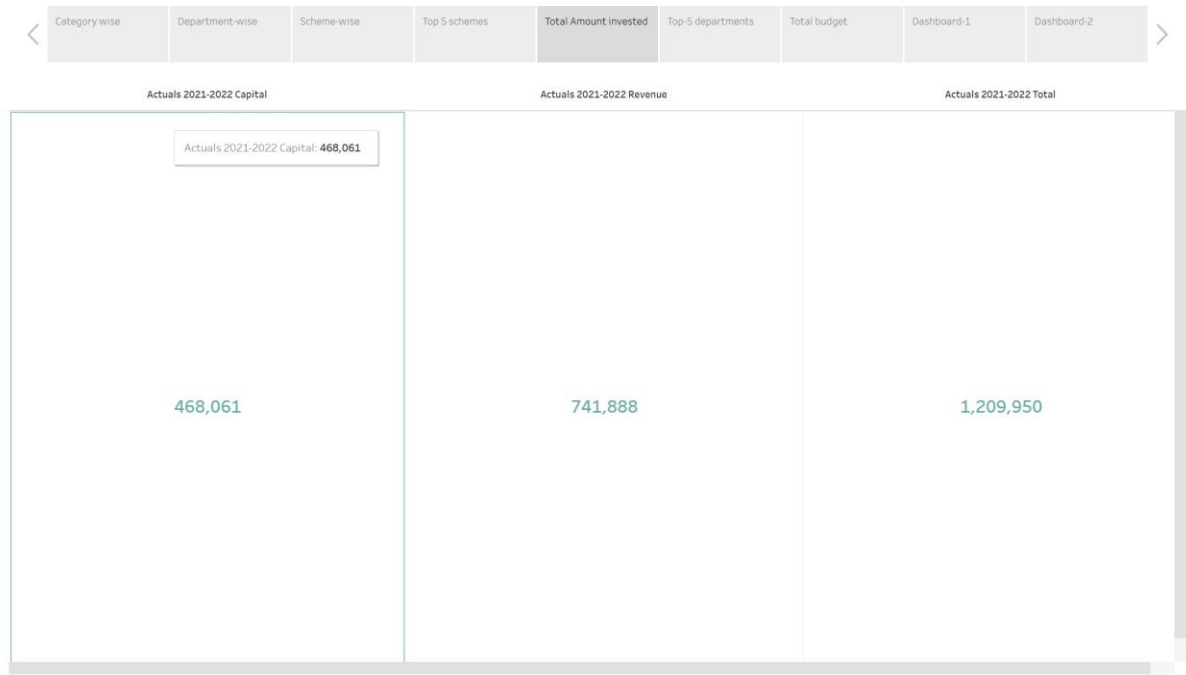
Story 1



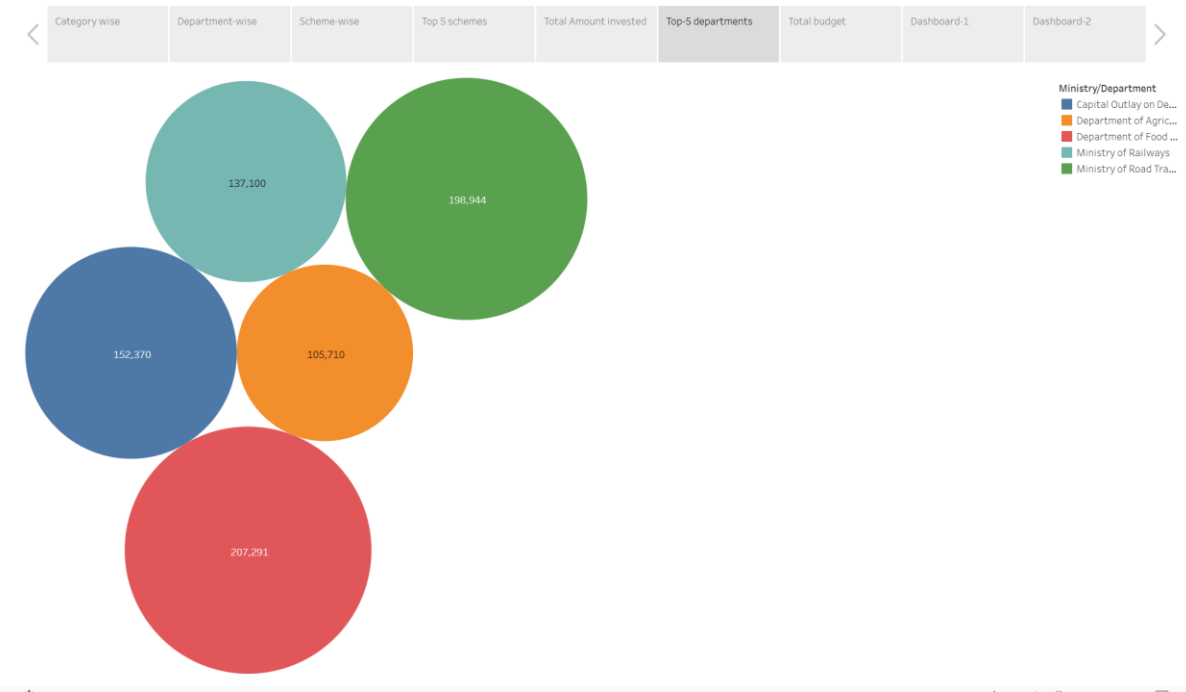
Story 1



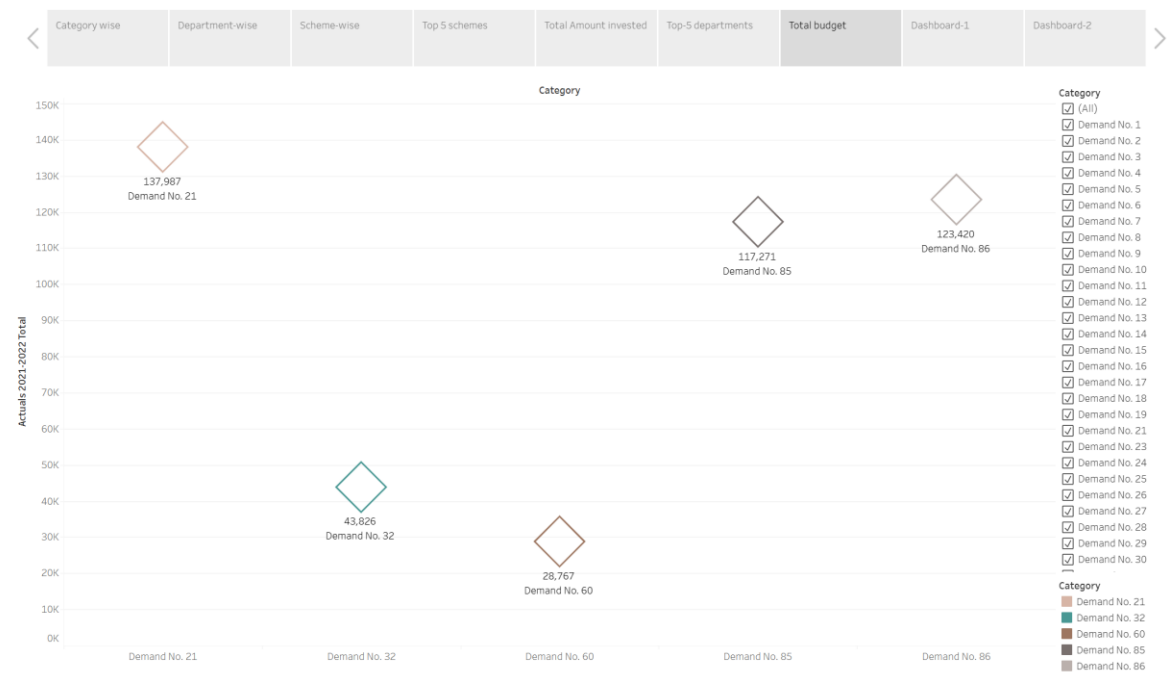
Story 1



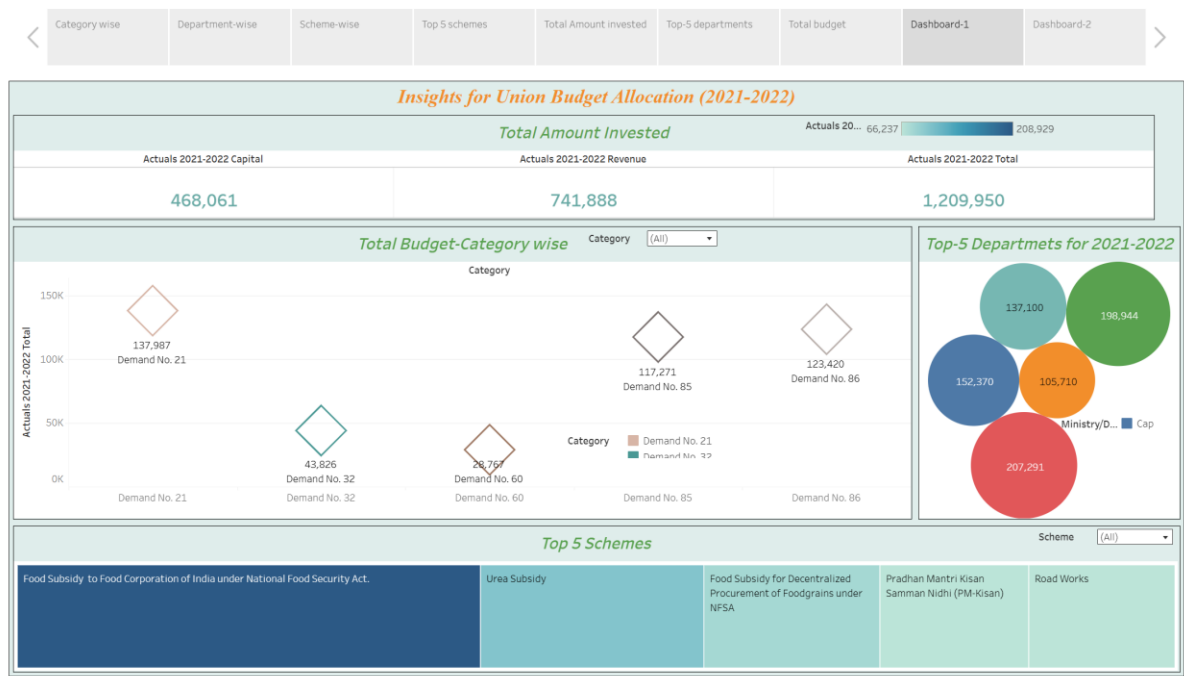
Story 1



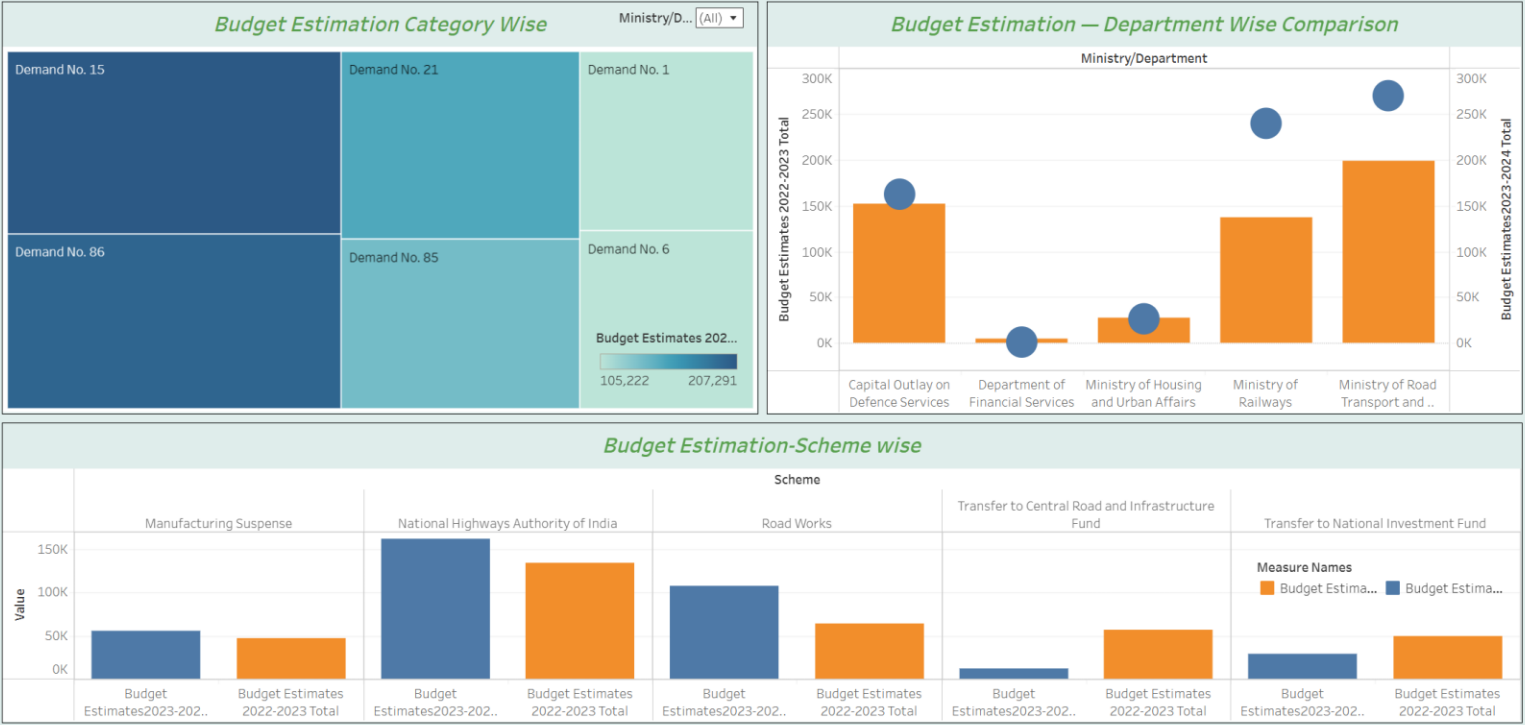
Story 1



Story 1



Insights of Union Budget Allocation(2022-2024)



ADVANTAGES & DISADVANTAGES

Advantages

- Provides a **clear and structured analysis** of Union Budget allocations across multiple financial years.
- Helps in **identifying trends, priorities, and sector-wise growth** in government spending.
- Interactive **Tableau dashboards and stories** make complex financial data easy to understand.
- Use of **SQL and database integration** improves data handling, accuracy, and performance.
- Web integration using **Flask** allows users to access dashboards through a user-friendly interface.
- Useful for **businesses, policymakers, and researchers** to support data-driven decisions

Disadvantages

- The analysis depends on the **availability and accuracy of historical budget data**.
- Limited to selected financial years (2021–22 to 2023–24), which may not capture long-term trends.
- Requires **technical knowledge of SQL, Tableau, and Flask** for development and maintenance.
- Real-time budget updates are not supported in the current implementation.

CONCLUSION

This project successfully analyzes the evolution of India's Union Budget allocations for the financial years 2021–22 to 2023–24, highlighting the government's focus on economic recovery, inclusive growth, and sustainable development. By combining data collection, SQL operations, Tableau visualizations, and interactive dashboards, the project transforms raw budget data into meaningful insights. The use of dashboards and data stories helps users easily understand allocation patterns across ministries, schemes, and categories. Overall, the project demonstrates how data analytics and visualization tools can effectively support transparency, better understanding of public finance, and informed decision-making.

FUTURE SCOPE

- Extend the analysis to **additional financial years** for long-term trend evaluation.
- Integrate **real-time budget and expenditure data** from government portals.
- Add **advanced analytics and forecasting models** to predict future budget allocations.
- Enhance dashboards with **AI-based insights and recommendations**.
- Improve the web application with **user authentication and role-based access**.
- Expand the project to include **state-wise budget analysis** for deeper regional insights

APPENDIX

Source code: <https://github.com/shruthitamaraana/Empower-India>